Sequential Sampling for Simulation-Test-Integrated Global Optimization in High Dimensions

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Abstract

Keywords:

1. Introduction

Simulation-Test Integrated Global Optimization (STIGO)

2. Review of Simulation-Test Integrated Global Optimization

The problem concerned by STIGO is to find the maximum point of a black-box function $\mathcal{P}(\boldsymbol{x})$ over the feasibility region $\boldsymbol{x} \in \mathbb{X} \subset \mathbb{R}^n$, i.e.,

$$\boldsymbol{x}^* = \text{argmax}_{\boldsymbol{x}\in\mathbb{X}} \mathcal{P}(\boldsymbol{x}) \tag{1}$$

, where the objective function value can be either observed from an experimental measurement with random noise $\varepsilon(\boldsymbol{x})$, or evaluated by calling a computational model $\mathcal{M}(\boldsymbol{x}, \boldsymbol{\theta})$, with $\boldsymbol{\theta}$ being a d-dimensional deterministic-but-unknown model parameters. In case it is observed from a physical test, the measurement error $\varepsilon(\boldsymbol{x})$ is assumed to be Gaussian white noise, i.e., $\varepsilon(\boldsymbol{x}) \equiv \varepsilon \sim \mathcal{N}(0, \sigma_{\text{n}}^2)$ with σ_{n}^2 being the variance; while in case it is computed with the computational model, a probability density $p_0(\boldsymbol{\theta})$, supported by $\mathbb{T} \subset \mathbb{R}^d$, is assumed and reflects the model uncertainty. The target of STIGO is to find the global maximum point of $\mathcal{P}(\boldsymbol{x})$, with as less costs of both experiments and computations (of model) as possible.

Within the STIBO framework, Bayesian optimal experimental design (BOED), physical test and measurement, as well as Bayesian model updating (BMU), are performed alternately. To be specific, for the $(t-1)$-th stage of STIBO, let denote by $\mathbf{x}_{1:t-1}$ the collections of Optimal experimental designs (OEDs) up to the $(t-1)$-th the stage, and $\mathbf{y}_{1:t-1}$ the collection of the corresponding experimental measurements. Denote by y_0 the initial reference objective value before implementing any physical test, which can be determined based on experience, or from the initial sample set, as will be discussed in numerical implementation of

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BOED. Further, let $p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1},\mathbf{y}_{1:t-1}\right)$ be the posterior density of the model parameters generated by the BMU procedure for the $(t-1)$-th stage. Then, for the t-th stage, the BOED procedure is initially implemented to achieve the OED $\mathbf{x}_t$ by maximizing the following utility function over $\mathbb{X}$:

$$\mathcal{U}_t\left(\boldsymbol{x}|\mathbf{x}_{1:t-1},\mathbf{y}_{1:t-1}\right)=\int_{\mathbb{T}}u_t\left(\boldsymbol{z}\right)p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1},\mathbf{y}_{1:t-1}\right)\mathrm{d}\boldsymbol{\theta} \tag{2}$$

, where $\boldsymbol{z}=(\boldsymbol{x},\boldsymbol{\theta})$ and

$$u_t\left(\boldsymbol{z}\right)=\left(\mathcal{M}\left(\boldsymbol{z}\right)-y_{\max}\right)\Phi\left(\frac{\mathcal{M}\left(\boldsymbol{z}\right)-y_{\max}}{\sigma_{\mathrm{n}}}\right)+\sigma_{\mathrm{n}}\phi\left(\frac{\mathcal{M}\left(\boldsymbol{z}\right)-y_{\max}}{\sigma_{\mathrm{n}}}\right) \tag{3}$$

, with $y_{\max}=\max\left(\mathbf{y}_{0:t-1}\right)$ being the reference value, computed to be the maximum measurements up to the $(t-1)$-th stage. Mathematically, the value of $\mathcal{U}_t$-function quantifies the expected improvement of the value $\mathcal{P}\left(\boldsymbol{x}\right)$ compared to $y_{\max}$, and thus its maximum point $\mathbf{x}_t$ can be explained at the point which is most informative to the global maximum point of $\mathcal{P}\left(\boldsymbol{x}\right)$. Given the measurement y_t is achieved by implementing the physical test at $\mathbf{x}_t$, the likelihood function of the measurement conditioning on $\boldsymbol{\theta}$ is formulated as:

$$\mathcal{L}\left(\mathbf{x}_t,\mathrm{y}_t|\boldsymbol{\theta}\right)=\phi\left(\mathcal{M}\left(\mathbf{x}_t,\boldsymbol{\theta}\right)-\mathrm{y}_t|0,\sigma_{\mathrm{n}}^2\right). \tag{4}$$

With the posterior density $p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1},\mathbf{y}_{1:t-1}\right)$ of the $(t-1)$-th stage as the prior density, the posterior density for BMU at the t-th stage is formulated as:

$$p_t\left(\boldsymbol{\theta}|\mathbf{x}_{1:t},\mathbf{y}_{1:t}\right)=\frac{\mathcal{L}\left(\mathbf{x}_t,\mathrm{y}_t|\boldsymbol{\theta}\right)p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1},\mathbf{y}_{1:t-1}\right)}{C_t} \tag{5}$$

, where

$$C_t=\int_{\mathbb{T}}\mathcal{L}\left(\mathbf{x}_t,\mathrm{y}_t|\boldsymbol{\theta}\right)p_{t-1}\left(\boldsymbol{\theta}|\mathbf{x}_{1:t-1},\mathbf{y}_{1:t-1}\right)\mathrm{d}\boldsymbol{\theta} \tag{6}$$

is the constant for normalizing the posterior density p_t, and is called model evidence.

The stopping condition of STIBO is formulated as:

$$\mathcal{U}_t\left(\mathbf{x}_t|\mathbf{x}_{1:t-1},\mathbf{y}_{1:t-1}\right)<\epsilon \tag{7}$$

, with ϵ being a σ_{n}-dependent threshold. In case Eq. (7) is satisfied, the expected improvement at most informative OED $\mathbf{x}_t$ is less than the small threshold ϵ, and then $\mathbf{x}_t$ or $\mathbf{x}_{t-1}$ is believed, with high credibility, as the global maximum point of $\mathcal{P}\left(\boldsymbol{x}\right)$.

The numerical challenges for the t-th stage include the estimation of the maximum point $\mathbf{x}_t$ of $\mathcal{U}_t$-function, the value of $\mathcal{U}_t$ at $\mathbf{x}_t$, as well as the posterior density p_t. In the next subsection, a sampling method based on adaptive parallel tempering is introduced for addressing the above challenges. For simplicity, in what

follows we drop the conditions of p_t and $\mathcal{U}_t$ and denote them as $p_t(\boldsymbol{\theta})$ and $\mathcal{U}_t(\boldsymbol{x})$ respectively.

3. Numerical Implementation of STIGO with Sequential Importance Sampling

3.1. Reformulation of BOED and BMU as sampling

As has been presented in Ref. [], u_t is non-negative and monotonically non-decreasing w.r.t. the model function $\mathcal{M}$. Inspired by the non-negativity, a joint density in the augmented space $\mathbb{Z} = \mathbb{X} \times \mathbb{T} \subset \mathbb{R}^{n+d}$ is defined as:

$$\pi_t(\boldsymbol{x}, \boldsymbol{\theta}) = \frac{u_t(\boldsymbol{x}, \boldsymbol{\theta}) f(\boldsymbol{x}) p_{t-1}(\boldsymbol{\theta})}{c_t} \tag{8}$$

, from which, integrating $\boldsymbol{\theta}$ out yields a marginal density formulated as:

$$h_t(\boldsymbol{x}) = \frac{\mathcal{U}_t(\boldsymbol{x}) f(\boldsymbol{x})}{c_t} \tag{9}$$

, where $f(\boldsymbol{x})$ refers to the density of uniform probability distribution supported by $\mathbb{X}$, and c_t is the corresponding normalizing constant formulated as:

$$c_t = \int_{\mathbb{X}\times\mathbb{T}} u_t(\boldsymbol{x}, \boldsymbol{\theta}) f(\boldsymbol{x}) p_{t-1}(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{\theta} = \int_{\mathbb{X}} \mathcal{U}_t(\boldsymbol{x}) f(\boldsymbol{x}) \mathrm{d}\boldsymbol{x}. \tag{10}$$

Specifically, for $t = 0$, π_0 is defined as:

$$\pi_0(\boldsymbol{x}, \boldsymbol{\theta}) = f(\boldsymbol{x}) p_0(\boldsymbol{\theta}) \tag{11}$$

, $u_0(\boldsymbol{z}) = 1$ and c_0=1.

With the formulation of Eq. (8) and Eq. (9) for BOED as well as Eq. (5) for BMU, Lemma 1 can be reached and plays as a foundation for efficient sampling. One can refer to Appendix Appendix A for mathematical proof. This lemma provides a theoretical foundation for generating sample set $\mathbf{T}_t$ following p_t starting from those $\mathbf{T}_{t-1}$ following p_{t-1}, and constructing $\mathbf{Z}_t$ following π_t starting from $\mathbf{Z}_{t-1}$ which follows exactly π_{t-1}.

Lemma 1. *The supports of the three densities p_t, π_t and h_t satisfy* $\mathrm{supp}(p_t) \subseteq \mathrm{supp}(p_{t-1})$, $\mathrm{supp}(\pi_t) \subseteq \mathrm{supp}(\pi_{t-1})$ *and* $\mathrm{supp}(h_t) \subseteq \mathrm{supp}(h_{t-1})$.

Further, Lemma 2 is obtained and proved in Appendix B. It indicates that, as long as the sample set $\mathbf{Z}_t$ follows π_t is generated, its marginal sample set $\mathbf{X}_t$ in the marginal space $\mathbb{X}$ can then be used for trivially searching the global maximum point of the utility function $\mathcal{U}_t$. With this conclusion, the challenging nested-optimization problem for BOED is transformed into a sampling problem defined in the augmented space $\mathbb{Z}$.

Lemma 2. *Given a sample set $\mathbf{Z}_t$ of z following the distribution π_t, the resultant marginal sample set $\mathbf{X}_t$ of x follows exactly the distribution h_t, and it further holds that:* $\arg\max_{x\in\mathbf{X}} \mathcal{U}_t(x) = \arg\max_{x\in\mathbf{X}} h_t(x)$. *In other words, among the marginal sample set $\mathbf{X}$, the sample with the highest density is exactly the global maximum of the utility function $\mathcal{U}_t$.*

3.2. Sequential Importance Sampling for BOED

Suppose a sample set $\mathbf{Z}_{t-1} = \left\{z_{t-1}^{(k)}\right\}_{k=1:N}$ following π_{t-1} is readily available. Specifically, for $t = 1$, $\mathbf{Z}_0$ can be generated by Latin Hypercube Sampling (LHS) following π_0, after evaluating the value of the model function $\mathcal{M}$ at these N sample points, y_0 can be empirically specified as, e.g., a specific sample quantile of the model response. The value of y_0 can then be substituted into Eq. (3) for formulating u_1 and then the density function π_1 with Eq. (8).

The key procedure for addressing BOED is to generate sample set $\mathbf{Z}_t$ of the same size with $\mathbf{Z}_{t-1}$, starting from $\mathbf{Z}_{t-1}$. To this purpose, we first reformulate Eq. (8) as:

$$\pi_t(z) = \frac{\tilde{u}_t(z)\,\pi_{t-1}(z)}{\tilde{c}_t} \tag{12}$$

, where $\tilde{u}_t(z) = u_t(z)/u_{t-1}(z)$ and $\tilde{c}_t = c_t/c_{t-1}$. As $\mathrm{supp}(\pi_t) \subseteq \mathrm{supp}(\pi_{t-1})$ holds, by analogy with Bayes' theorem, π_t can be interpreted as a posterior density given π_{t-1} as the prior density and $\tilde{u}_t$ as the likelihood function. Further, an Artificial Intermediate Density (AID) is formulated as:

$$\pi_{t,j}(z) = \frac{[\tilde{u}(z)]^{\gamma_j}\,\pi_{t-1}(z)}{\tilde{c}_{t,j}} \tag{13}$$

, where $\gamma_j \in [0,1]$. It is obvious to seen that, given $\gamma_j = 0$, Eq. (13) equals exactly π_{t-1}; while given $\gamma_j = 1$, it is exactly π_t. In case a considerable divergence exist between π_{t-1} and π_t, it is usually challenging to draw samples following π_t starting from the samples of π_{t-1} via resampling or any MCMC sampling schemes. In this case, the AIDs formulated with Eq. (13) can be served as a set of intermediate densities for bridging π_{t-1} and π_t, allowing for generating samples sequentially by designing a set of inverse temperatures $0 = \gamma_0 < \gamma_1 < \cdots < \gamma_m = 1$. Specifically, initialized with $\gamma_0 = 0$ and $\mathbf{Z}_{t,0} = \mathbf{Z}_{t-1}$, and given γ_{j-1} fixed and the corresponding sample set $\mathbf{Z}_{t,j-1}$ following $\pi_{t,j-1}$ readily generated, two key procedures are required for the j-th tempering stage. First, the value of γ_j needs to be specified controlling the ratio between $\tilde{c}_{t,j}$ and $\tilde{c}_{t,j-1}$. Second, a sample population $\mathbf{Z}_{t,j}$ following $\pi_{t,j}$ needs to be generated starting from $\mathbf{Z}_{t,j-1}$.

Based on Eq. (13), the ratio between $\tilde{c}_{t,j}$ and $\tilde{c}_{t,j-1}$ is expressed as:

$$\frac{\tilde{c}_{t,j}}{\tilde{c}_{t,j-1}} = \int_{\mathbb{X}\times\mathbb{T}} [\tilde{u}_t(z)]^{\gamma_j-\gamma_{j-1}}\,\pi_{t,j-1}(z)\mathrm{d}z. \tag{14}$$

Then, for each sample $\boldsymbol{z}_{t,j-1}^{(k)} \in \mathbf{Z}_{t,j-1} \sim \pi_{t;j-1}$, a weight can be defined as:

$$w^{(k)}\left(\gamma_j\right) = \left[\tilde{u}_t\left(\boldsymbol{z}_{t,j-1}^{(k)}\right)\right]^{\gamma_j - \gamma_{j-1}}. \tag{15}$$

As $\operatorname{supp}\left(u_t\right) \subseteq \operatorname{supp}\left(u_{t-1}\right)$ and $\operatorname{supp}\left(\pi_{t,j-1}\right) \subseteq \operatorname{supp}\left(\pi_{t-1}\right) \subseteq \operatorname{supp}\left(u_{t-1}\right)$ holds, the values of u_{t-1}, and thus $\tilde{u}_t$, for all samples have finite value, and then all weight samples $w^{(k)}\left(\gamma_j\right)$ are well defined without existence of infinity value. As the sample mean $N^{-1}\sum_{k=1}^{N} w^{(k)}\left(\gamma_j\right)$, is utilized as a MC estimator for estimating $\tilde{c}_{t,j}/\tilde{c}_{t,j-1}$, as indicated by Eq. (14), it is expected that the weight samples does not exhibit excessively high variability, so that the variability of this MC estimator can be reasonably controlled. Practically, we specify γ_j by controlling the Coefficient of Variation (CoV) of the weigh samples, i.e., $\operatorname{CoV}\left[w^{(k)}\left(\gamma_j\right)\right]$, under a pre-specified threshold ζ, which means, in case $\operatorname{CoV}\left[w^{(k)}\left(1\right)\right] < \zeta$, let $\gamma_j = 1$; other wise compute γ_j by:

$$\gamma_j = \operatorname{argmin}_{\gamma \in (\gamma_{j-1}, 1]} \left| \operatorname{CoV}\left[w^{(k)}\left(\gamma\right)\right] - \zeta \right| \tag{16}$$

With γ_j being determined, we denote $w_j^{(k)} = w^{(k)}\left(\gamma_j\right)$ and the normalized weights $\tilde{w}_j^{(k)} = w_j^{(k)} / \sum_{j=1}^{N} w_j^{(k)}$. Then, the discrete probability distribution law:

$$\mathbb{P}\left[\boldsymbol{z} = \boldsymbol{z}_{t,j-1}^{(k)}\right] = \tilde{w}_j^{(k)}, \quad \forall \boldsymbol{z}_{t,j-1}^{(k)} \in \mathbf{Z}_{t,j-1} \tag{17}$$

is an approximation of the continuous distribution $\pi_{t,j}$. Following this distribution law, a candidate sample population $\bar{\mathbf{Z}}_{t,j} = \left\{\bar{\boldsymbol{z}}_{t,j}^{(k)}\right\}_{k=1:N}$ can be generated with replacement, which follows exactly $\pi_{t,j}$ but inevitably includes some duplicate samples. This procedure is called resampling. The covariance of this discrete distribution can be computed as:

$$\Sigma_{t,j} = \sum_{k=1}^{N} \tilde{w}_j^{(k)} \left(\boldsymbol{z}_{t,j-1}^{(k)} - \sum_{k=1}^{N} \tilde{w}_j^{(k)} \boldsymbol{z}_{t,j-1}^{(k)}\right)^{\top} \left(\boldsymbol{z}_{t,j-1}^{(k)} - \sum_{k=1}^{N} \tilde{w}_j^{(k)} \boldsymbol{z}_{t,j-1}^{(k)}\right). \tag{18}$$

To alleviate sample degradation, following the traditional treatment of a SMC procedure (see e.g., [1, 2]) , it is suggested to grow a Markov chain, with two states, by setting each sample $\bar{\boldsymbol{z}}_{t,j}^{(k)} \in \bar{\mathbf{Z}}_{t,j}$ as the initial state, and then collect the new state of each chain to form the sample population $\mathbf{Z}_{t,j}$. This procedure can be realized with any MCMC sampling schemes such as the Metropolis-Hasting (MH) algorithm, Hamiltonian Monte Carlo (HMC), and Metropolis-Adjusted Langevin Algorithm (MALA). The latter two usually show much higher acceptance rate, especially for high-dimensional cases, but both require evaluation of the gradients of stationary density. Here, we take the MH algorithm as an example. With $\bar{\boldsymbol{z}}_{t,j}^{(k)}$ as an initial state, the following two procedures are implemented to generate the second state $\boldsymbol{z}_{t,j}^{(k)}$.

i. Generate a candidate sample $\boldsymbol{z}_{\text{cand}}$ following a proposal density, which, for example, can be formulated

as the Gaussian density with mean $\bar{\boldsymbol{z}}_{t,j}^{(k)}$ and covariance $\eta^2\Sigma_{t,j}$, where η is a scaling parameter which is suggested to be 0.2 following Ref. [1].

ii. Accept $\boldsymbol{z}_{\text{cand}}$ as $\boldsymbol{z}_{t,j}^{(k)}$ with a probability $\alpha = \min\left(1, \pi_t\left(\boldsymbol{z}_{\text{cand}}\right)/\pi_t\left(\bar{\boldsymbol{z}}_{t,j}^{(k)}\right)\right)$; otherwise, set $\boldsymbol{z}_{t,j}^{(k)} = \bar{\boldsymbol{z}}_{t,j}^{(k)}$.

With the sample population $\mathbf{Z}_{t,j}$ being generated with the above Resampling-then-Metropolis-Hasting (RMH) scheme, the ratio $\tilde{c}_{t,j}/\tilde{c}_{t,j-1}$ can then be estimated by either [1]

$$\frac{\tilde{c}_{t,j}}{\tilde{c}_{t,j-1}} \cong \frac{1}{N}\sum_{k=1}^{N}\left[\tilde{u}_t\left(\boldsymbol{z}_{t,j-1}^{(k)}\right)\right]^{\gamma_j-\gamma_{j-1}} \tag{19a}$$

, or [3]

$$\frac{\tilde{c}_{t,j}}{\tilde{c}_{t,j-1}} \cong \frac{\sum_{k=1}^{N}\left[\tilde{u}_t\left(\boldsymbol{z}_{t,j-1}^{(k)}\right)\right]^{\frac{1}{2}(\gamma_j-\gamma_{j-1})}}{\sum_{k=1}^{N}\left[\tilde{u}_t\left(\boldsymbol{z}_{t,j}^{(k)}\right)\right]^{\frac{1}{2}(\gamma_{j-1}-\gamma_j)}}. \tag{19b}$$

The value of this ratio is indeed not required for BOED.

Suppose now for the j-th tempering stage, γ_j is specified as 1, then let $m = j$, and $\mathbf{Z}_t = \mathbf{Z}_{t,m}$ follows exactly the distribution π_t. The marginal density $h(\boldsymbol{x})$ can be formulated as:

$$h_t(\boldsymbol{x}) \propto \int_{\mathbb{X}}\int_{\mathbb{T}} \frac{\pi_t(\boldsymbol{x},\boldsymbol{\theta})}{\pi_t(\boldsymbol{x}',\boldsymbol{\theta})}\pi_t(\boldsymbol{x}',\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta}\mathrm{d}\boldsymbol{x}' \tag{20}$$

, where $\boldsymbol{x}'$ is an independent replication of $\boldsymbol{x}$. Thus, searching the global maximum of $h_t(\boldsymbol{x})$ is equivalent to maximize the following unbiased estimator:

$$\hat{h}(\boldsymbol{x}) = \frac{1}{N}\sum_{k=1}^{N}\frac{\pi_t\left(\boldsymbol{x},\boldsymbol{\theta}_t^{(k)}\right)}{\pi_t\left(\boldsymbol{x}_t^{(k)},\boldsymbol{\theta}_t^{(k)}\right)} \tag{21}$$

With the initial point $\boldsymbol{x}_{t,0}$ specified as the one, out of $\mathbf{Z}_t$, with the highest joint density, i.e.,

$$\boldsymbol{x}_{t,0} = \text{argmax}_{(\boldsymbol{x},\boldsymbol{\theta})\in\mathbf{Z}_t}\pi_t(\boldsymbol{x},\boldsymbol{\theta}) \tag{22}$$

, a gradient-base optimization can then be used for searching the OED $\mathbf{x}_t$.

3.3. Sequential Sampling for BMU

Let $\mathbf{T}_{t-1}$ denote a readily produced sample population following the prior distribution p_{t-1}. Specifically, for $t = 1$, $\mathbf{T}_0$ can be set as the marginal sample population of $\mathbf{Z}_0$; otherwise, it is produced by the BMU procedure of the $(t-1)$-th stage. Inevitably, large discrepancies will arise between p_{t-1} and p_t. An AID bridging the transition from the prior p_{t-1} to the posterior needs to be constructed. Following the treatment

of TMCMC in Ref. [1], the AID for this purpose is formulated as:

$$p_{t,j}(\boldsymbol{\theta}) = \frac{[\mathcal{L}(\mathbf{x}_t, \mathrm{y}_t|\boldsymbol{\theta})]^{\beta_j} p_{t-1}(\boldsymbol{\theta})}{C_{t,j}} \tag{23}$$

, where $\beta_j \in [0,1]$. Similarly, with β_j=0, $p_{t,j} = p_{t-1}$; and with β_j=1, $p_{t,j} = p_t$ exactly. Thereof, one can

3.4. Summary of the sampling-based STIBO algorithm

4. Benchmark Studies

4.1. An illustrative numerical example

4.2. Conclusions and Prospects

extension to problem with also extension to reliability sensitivity analysis: in two settings extension to RBDO

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Appendix A. Proof of Lemma 1

Appendix B. Proof of Lemma 2

$$h_t(\boldsymbol{x}) = \int_{\mathbb{X}} \pi_t(\boldsymbol{x}, \boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta} \tag{B.1}$$

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